Linear Algebra I Recitation Class: Week 10

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Throughout these notes, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ denote the sets of rational, real, and complex numbers, respectively. V is a vector space over a field F. If you are not yet familiar with fields, you may keep the examples $F = \mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$ in mind. If T is a linear transformation, we use the notation $\ker T := N(T)$ and $\operatorname{im} T := R(T)$.

Let R be a commutative ring (for example, $R = \mathbb{Z}$). We say a square matrix $A \in M_{n\times n}(R)$ is **invertible over** R if A is an invertible matrix and its inverse satisfies $A^{-1} \in M_{n\times n}(R)$. For example, if $R = \mathbb{Z}$, then $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is invertible over $\mathbb{Z}$ with the inverse $\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$, while $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ is not invertible over $\mathbb{Z}$ since its inverse matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1/2 \end{pmatrix}$ is not defined over $\mathbb{Z}$ ($1/2 \notin \mathbb{Z}$). We define the set:

$$\mathrm{GL}_m(R) := \{P \in M_{m\times m}(R) \mid P \text{ is invertible over } R\}.$$

Using elementary row/column operations, for any $A \in M_{m\times n}(\mathbb{Q})$, there exist $P \in \mathrm{GL}_n(\mathbb{Q})$ and $Q \in \mathrm{GL}_m(\mathbb{Q})$ such that

$$PAQ = \begin{pmatrix} 1 & & & & & & \\ & 1 & & & & & \\ & & \ddots & & & & \\ & & & 1 & & & \\ & & & & 0 & & \\ & & & & & \ddots & \\ & & & & & & 0 \end{pmatrix} =: D,$$

where the rank of A equals the number of 1s in PAQ. Note that $P^{-1}DQ^{-1} = A$ and $P^{-1} \in \mathrm{GL}_n(\mathbb{Q})$, $Q \in \mathrm{GL}_m(\mathbb{Q})$; therefore, A can be recovered using matrices defined over $\mathbb{Q}$.

We would like to ask if we have a similar result for matrices $A \in M_{m\times n}(\mathbb{Z})$. However, this does not hold in general: there is no way to express $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ in the form $P \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} Q$ with P, Q invertible over $\mathbb{Z}$. The main reason is that when performing elementary operations of types II and III on matrices over $\mathbb{Q}$, we are allowed to multiply a row/column by a nonzero scalar (in $\mathbb{Q}$) and add a scalar multiple of one row/column to another. Because $\mathbb{Q}$ is a field, the inverses of the corresponding elementary matrices are also defined over $\mathbb{Q}$. While this holds for $\mathbb{Q}$, it fails for $\mathbb{Z}$: for instance, $2 \in \mathbb{Z}$, but its multiplicative inverse $1/2 \notin \mathbb{Z}$. Therefore, in order to ensure the elementary operations behave "nicely" for matrices over $\mathbb{Z}$, we need to modify the definition of elementary operations:

Definition 1. Let A be an $m \times n$ matrix defined over $\mathbb{Z}$. An **elementary row/column operation over $\mathbb{Z}$** is any of the following operations:

1. Interchanging any two rows/columns of A.
2. Multiplying any row/column by ± 1.

3. Adding any $\mathbb{Z}$-**scalar** multiple of a row/column of A to another row/column.

The **elementary matrices over** $\mathbb{Z}$ are the matrices corresponding to elementary row/column operations over $\mathbb{Z}$. One can easily that they are all invertible over $\mathbb{Z}$.

With these restricted operations, we can obtain a result similar to the one over $\mathbb{Q}$:

Theorem 2 (Existence of the Smith Normal Form)**.** *Given any $A \in M_{m\times n}(\mathbb{Z})$, there exist invertible matrices $P \in M_{m\times m}(\mathbb{Z})$ and $Q \in M_{m\times n}(\mathbb{Z})$ such that*

$$PAQ = \begin{pmatrix} d_1 & & & & & & \\ & d_2 & & & & & \\ & & \ddots & & & & \\ & & & d_k & & & \\ & & & & 0 & & \\ & & & & & \ddots & \\ & & & & & & 0 \end{pmatrix} =: D$$

with $d_1, \ldots, d_k$ non-zero and $d_1 \mid d_2 \mid \cdots \mid d_k$.

Remark. It can be proved that D is unique (up to a sign of ± 1 on each entry); therefore, it makes sense to say D is *the* **Smith normal form** of A. The proof of uniqueness is beyond our scope right now, but hopefully, we will return to it in the near future.

Proof. (Based on notes for Shin-Yao Jow's Advanced Linear Algebra Course) If $A = 0$ there is nothing to do, so we may, after switching rows and columns, assume that A has a $(1,1)$ entry $a \neq 0$. We can use elementary operations to decrease the absolute value of the $(1,1)$ entry in the following cases:

Case 1. If a does not divide some entry in the first row, after switching columns we may assume that a does not divide the $(1,2)$ entry b. Write $b = aq + r$ with $r \neq 0$. Subtract q times column 1 from column 2 to make the $(1,2)$ entry r. Switch columns 1 and 2 so that r appears in the $(1,1)$ position.

Case 2. If a does not divide some entry in the first column, we can use a similar process as in Case 1 to decrease the value of the $(1,1)$ entry.

Keep applying these two cases until they no longer apply, i.e., when the $(1,1)$ entry divides every entry in the first row and the first column. Then we can use elementary operations to annihilate every entry other than the $(1,1)$ entry in the first row and the first column.

Case 3. Suppose every entry other than the $(1,1)$ entry in the first row and the first column is 0, and the $(1,1)$ entry $a \neq 0$ does not divide some entry of the matrix. After switching columns and rows, we may assume that a does not divide the $(2,2)$ entry. Add row 2 to row 1, and we are back in Case 1.

Repeat the entire process (Cases 1, 2, and 3) until we can't anymore, i.e., when the matrix is of the form $\begin{pmatrix} a & 0 \\ 0 & B \end{pmatrix}$ where $B \in M_{m-1,n-1}(R)$ and a divides every entry of B. Now repeat the whole process on B, and we are done by induction. $\square$

The proof can be difficult to understand initially, but it becomes extremely straightforward once you work through some concrete examples. Let A be a 3×3 matrix over $\mathbb{Z}$:

$$A = \begin{pmatrix} 2 & 4 & 4 \\ -3 & -2 & -5 \\ 5 & 6 & 9 \end{pmatrix}$$

The $(1,1)$ entry is 2. It divides everything in row 1, but it does not divide the -3 in column 1. We write -3 using division with remainder: $-3 = 2(-2) + 1$.

To replace the -3 with the remainder 1, we add $2 \times R_1$ to R_2:

$$\xrightarrow{R_2 \leftarrow R_2 + 2R_1 \ (1)} \quad \begin{pmatrix} 2 & 4 & 4 \\ 1 & 6 & 3 \\ 5 & 6 & 9 \end{pmatrix}$$

Now, swap R_1 and R_2 to place the strictly smaller remainder into the $(1,1)$ position:

$$\xrightarrow{R_1 \leftrightarrow R_2 \ (2)} \quad \begin{pmatrix} 1 & 6 & 3 \\ 2 & 4 & 4 \\ 5 & 6 & 9 \end{pmatrix}$$

Because the $(1,1)$ entry is now 1, it divides every entry in its row and column. We use it to zero out the rest of the first column:

$$\begin{array}{l} R_2 \leftarrow R_2 - 2R_1 \ (3) \\ R_3 \leftarrow R_3 - 5R_1 \ (4) \end{array} \longrightarrow \quad \begin{pmatrix} 1 & 6 & 3 \\ 0 & -8 & -2 \\ 0 & -24 & -6 \end{pmatrix}$$

Next, zero out the rest of the first row using column operations:

$$\begin{array}{l} C_2 \leftarrow C_2 - 6C_1 \ (5) \\ C_3 \leftarrow C_3 - 3C_1 \ (6) \end{array} \longrightarrow \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & -8 & -2 \\ 0 & -24 & -6 \end{pmatrix}$$

The first row and column are clear. We now isolate the 2×2 block $B = \begin{pmatrix} -8 & -2 \\ -24 & -6 \end{pmatrix}$ and repeat the process for the $(2,2)$ position.

First, swap C_2 and C_3 to place the element with the smallest absolute value into the pivot position:

$$\xrightarrow{C_2 \leftrightarrow C_3 \ (7)} \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & -8 \\ 0 & -6 & -24 \end{pmatrix}$$

We can multiply R_2 by -1 (a unit in $\mathbb{Z}$) to make the pivot positive for convenience:

$$\xrightarrow{R_2 \leftarrow -R_2 \ (8)} \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 8 \\ 0 & -6 & -24 \end{pmatrix}$$

The new pivot, 2, already divides everything in its row and column. We annihilate the rest of column 2 and row 2:

$$\xrightarrow{R_3 \leftarrow R_3 + 3R_2 \ (9)} \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 8 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{C_3 \leftarrow C_3 - 4C_2 \ (10)} \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

The elementary matrices corresponding to each elementary operation are:

$$(1)\ E_1 = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ left.} \qquad (7)\ E_7 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \text{ right.}$$

$$(2)\ E_2 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ left.} \qquad (8)\ E_8 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ left.}$$

$$(3{+}4)\ E_{3+4} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -5 & 0 & 1 \end{pmatrix} \text{ left.} \qquad (9)\ E_9 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix} \text{ left.}$$

$$(5{+}6)\ E_{5+6} = \begin{pmatrix} 1 & -6 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ right.} \qquad (10)\ E_{10} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{pmatrix} \text{ right.}$$

Therefore,

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} = E_9 E_8 E_{3+4} E_2 E_1 A E_{5+6} E_7 E_{10} = \begin{pmatrix} 2 & 1 & 0 \\ 3 & 2 & 0 \\ -1 & 1 & 1 \end{pmatrix} A \begin{pmatrix} 1 & -3 & 6 \\ 0 & 0 & 1 \\ 0 & 1 & -4 \end{pmatrix} =: PAQ.$$

P and Q are invertible matrices over $\mathbb{Z}$ since they are products of elementary matrices defined over $\mathbb{Z}$.

Let's do another example. Let M be a 2×4 matrix over $\mathbb{Z}$:

$$M = \begin{pmatrix} 2 & 4 & -2 & 6 \\ 4 & 11 & -1 & 14 \end{pmatrix}$$

The $(1,1)$ entry is 2, which divides the $(2,1)$ entry 4. We can use row operations to directly eliminate the 4:

$$\xrightarrow{R_2 \leftarrow R_2 - 2R_1} \quad \begin{pmatrix} 2 & 4 & -2 & 6 \\ 0 & 3 & 3 & 2 \end{pmatrix}$$

The $(1,1)$ entry 2 also divides every other entry in its row. We use column operations to eliminate them:

$$\begin{array}{c} C_2 \leftarrow C_2 - 2C_1 \\ C_3 \leftarrow C_3 + C_1 \\ C_4 \leftarrow C_4 - 3C_1 \\ \longrightarrow \end{array} \quad \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 3 & 2 \end{pmatrix}$$

The first row and column are cleared, but our pivot 2 does NOT divide all entries in the remaining submatrix. We must bring an offending entry into the first row by adding the second row to the first:

$$\xrightarrow{R_1 \leftarrow R_1 + R_2} \quad \begin{pmatrix} 2 & 3 & 3 & 2 \\ 0 & 3 & 3 & 2 \end{pmatrix}$$

We subtract $1 \times C_1$ from C_2 to get the remainder:

$$\xrightarrow{C_2 \leftarrow C_2 - C_1} \quad \begin{pmatrix} 2 & 1 & 3 & 2 \\ 0 & 3 & 3 & 2 \end{pmatrix}$$

Swap C_1 and C_2 to put the smaller remainder into the pivot position:

$$\xrightarrow{C_1 \leftrightarrow C_2} \quad \begin{pmatrix} 1 & 2 & 3 & 2 \\ 3 & 0 & 3 & 2 \end{pmatrix}$$

Our pivot is now 1, which trivially divides everything. We clear the first column:

$$\xrightarrow{R_2 \leftarrow R_2 - 3R_1} \quad \begin{pmatrix} 1 & 2 & 3 & 2 \\ 0 & -6 & -6 & -4 \end{pmatrix}$$

And clear the first row:

$$\begin{array}{c} C_2 \leftarrow C_2 - 2C_1 \\ C_3 \leftarrow C_3 - 3C_1 \\ C_4 \leftarrow C_4 - 2C_1 \\ \longrightarrow \end{array} \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -6 & -6 & -4 \end{pmatrix}$$

We now look at the remaining 1×3 submatrix. To make things simpler, we move the element with the smallest absolute value (-4) to the pivot position and make it positive:

$$\xrightarrow{C_2 \leftrightarrow C_4} \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -4 & -6 & -6 \end{pmatrix}$$

$$\xrightarrow{C_2 \leftarrow -C_2} \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 4 & -6 & -6 \end{pmatrix}$$

The pivot 4 does not divide -6. Using division, $-6 = 4(-1) - 2$. We add C_2 to C_3:

$$\xrightarrow{C_3 \leftarrow C_3 + C_2} \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 4 & -2 & -6 \end{pmatrix}$$

Swap C_2 and C_3, then negate C_2 to get a positive pivot of 2:

$$\xrightarrow{C_2 \leftrightarrow C_3} \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 4 & -6 \end{pmatrix}$$

$$\xrightarrow{C_2 \leftarrow -C_2} \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 4 & -6 \end{pmatrix}$$

Finally, 2 divides 4 and -6. We clear the rest of the row:

$$\begin{array}{l} C_3 \leftarrow C_3 - 2C_2 \\ C_4 \leftarrow C_4 + 3C_2 \end{array} \xrightarrow{\qquad\qquad} \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \end{pmatrix}.$$

Exercise A. Trace the elementary matrices corresponding to each elementary operation above and find $P \in \mathrm{GL}_2(\mathbb{Z})$ and $Q \in \mathrm{GL}_4(\mathbb{Z})$ such that $PMQ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \end{pmatrix}$.

Smith normal forms are useful for determining if a system of linear equations over $\mathbb{Z}$ has a solution in $\mathbb{Z}$. If the answer is positive, it can also be used to find *all* solutions in $\mathbb{Z}$. We will talk about this more in the next recitation class. Another important application is classifying all **finitely generated abelian groups**, or more generally, all **finitely generated modules over a PID** (whatever that means), which gives an elegant approach to proving the existence of Jordan canonical forms and rational canonical forms.

Example 3. Let G be an abelian group defined by the relations

$$G = \left\langle s_1, s_2, \cdots, s_n \;\middle|\; s_i s_j = s_j s_i, \begin{cases} s_1^{a_{11}} s_2^{a_{12}} \cdots s_n^{a_{1n}} = 1 \\ s_1^{a_{21}} s_2^{a_{22}} \cdots s_n^{a_{2n}} = 1 \\ \qquad\qquad \vdots \\ s_1^{a_{m1}} s_2^{a_{m2}} \cdots s_n^{a_{mn}} = 1 \end{cases} \right\rangle.$$

Then G is a finitely generated abelian group. Let $A = (a_{ij}) \in M_{m \times n}(\mathbb{Z})$. If the Smith normal form of A is $\mathrm{diag}(d_1, d_2, \cdots, d_k, 0, \cdots, 0)$, then

$$G \cong \mathbb{Z}/d_1\mathbb{Z} \oplus \mathbb{Z}/d_2\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/d_k\mathbb{Z} \oplus \mathbb{Z}^{n-k}.$$

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